

S.No. : 346

NEC 4201

No. of Printed Pages : 06

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43401

Roll
No.

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B. Tech. Examination, 2026

(Even Semester)

BASIC ELECTRONICS ENGINEERING

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

- Draw the VI characteristics of an ideal diode.
- Write down the diode current equation.
- For $\beta = 100$ find the value of α .
- Write down the biasing condition of a BJT for saturation mode.

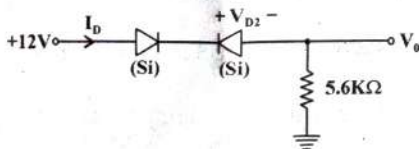
[P. T. O.]

- (e) Why FET is called voltage controlled device?
- (f) Draw the transfer characteristics for an n-channel JFET.
- (g) How to obtain average circuit from voltage summer?
- (h) Write down the 1's and 2's complement of $(10000)_2$.

SECTION - B

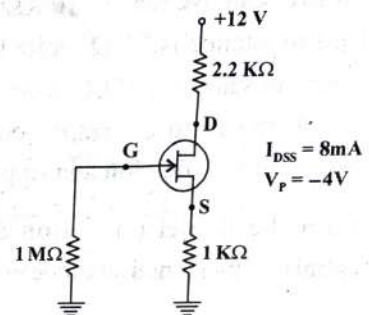
2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Explain the working of PN junction diode in forward biased condition. Calculate I_D , V_{D2} and V_0 for the following circuit :



- (b) What is a rectifier? Explain the working of centre-tapped full wave rectifier. Also draw its output waveform.

- (c) For the following circuit, calculate V_{GSQ} , I_{DQ} , V_{DS} , V_S and V_D :

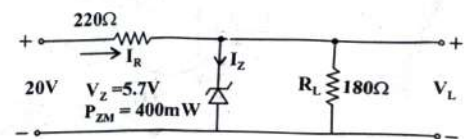


- (d) What are the universal gates? Draw the equivalent circuit of NOT, AND, OR and XOR gates using NOR gates only.

SECTION - C

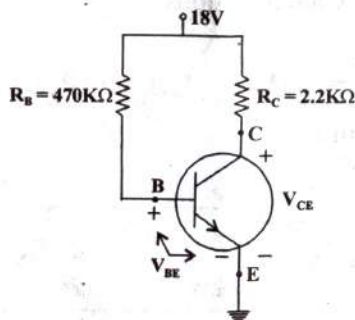
Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Determine V_L , I_L , I_Z , I_R and P_{ZM} for the following circuit :

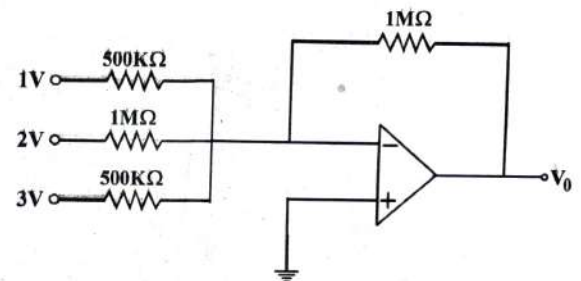


[P.T.O.]

- (b) A half wave rectifier circuit is supplied from a 230 V, 50 Hz supply with a step down ratio of 3 : 1 to a resistive load of 10 K Ω . The diode forward resistance is 75 Ω while transformer secondary resistance is 10 Ω . Calculate maximum, average, RMS value of current, d.c. output voltage, efficiency of rectification and ripple factor.
- (c) Explain the depletion region and barrier potential of a p-n junction diode.
4. (a) Define thermal runaway problem in BJT. Also derive the relationship between α and β .
- (b) Explain the construction and working of n-channel JFET.
- (c) For the following circuit, calculate I_C , V_{CE} , V_B , V_C and V_{BC} :



5. (a) Derive the expression of the gain for inverting summing amplifier.
- (b) Calculate the output voltage for the following circuit :



- (c) Perform $(27)_{10} - (10)_{10}$ using 1's and 2's complement.
6. (a) Why silicon diode is more preferred than germanium diode. The reverse saturation current of a silicon diode is 3 nA at 27 $^{\circ}\text{C}$. Find the reverse saturation current at 82 $^{\circ}\text{C}$.
- (b) Define base-width modulation in BJT. Also draw the input and output characteristics of a BJT in the CE configuration.
- (c) Convert the following :
- (i) $(23.6)_{10} = (?)_2$

(ii) $(10110111.11)_2 = (?)_8$

(iii) $(24)_{10} = (?)_{\text{Excess-3}}$

(iv) $(110101)_2 = (?)_{\text{gray}}$
